

Designing Building Blocks for Stacked Polyphase Bridge Converters in High-Voltage Traction Inverters

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Abstract—Stacked polyphase bridge (SPB) converters divide a high traction-battery voltage among series-connected low-voltage three-phase submodules, so that the transistor voltage class becomes a design variable rather than a consequence of the battery voltage. This paper presents a design procedure for the phase-leg building block of a 1200 V, 150 kW SPB traction inverter, which proceeds from the system specification to the voltage class, the transistor, the layout and gate-drive concept, the number of paralleled devices, the dc-link capacitors and finally the parasitic, loss and thermal verification. A device time constant derived for the SPB selects 100 V GaN transistors (EPC2361) and 16 submodules of 75 V among off-the-shelf devices of the 100–650 V classes; a star-connected driver per pair of transistors fixes the layout, and loss and driver count set two devices per switch. The resulting 75 V, 100 A_{rms} block on a 30 × 28 mm four-layer board reaches 0.374 nH per commutation cell and equal 1.7 nH gate loops, with an MLCC dc link sized by ESR- and ESL-based current sharing. The analysis further shows that the switching harmonics of the dc-link current add one third to the dc copper loss of the board, that the classical overlap loss formula is valid for a GaN turn-off only above a critical gate resistance, and that the board copper loss rather than the transistors sets the thermal design. At 50 kHz the block reaches 99.53 % efficiency with a junction temperature of 72 °C on a 60 °C cold plate.

Index Terms—Gallium nitride, dc-link capacitor, paralleling, parasitic inductance, PCB layout, stacked polyphase bridge, thermal design, traction inverter.

I. INTRODUCTION

Raising the battery voltage of electric vehicles to 800 V and beyond reduces the current and the charging time, but it forces a two-level traction inverter to 1200 V or higher-rated devices, steep voltage steps at the machine terminals and bulky dc-link capacitors. The stacked polyphase bridge (SPB) converter [1], [2] offers an alternative: N conventional three-phase bridges are connected in series on the dc side, each with its own dc-link capacitor and each feeding one three-phase winding set of a multiphase machine. Every submodule sees only V_{bus}/N , so low-voltage transistors with a far better on-resistance and switching figure of merit can be used, the submodule capacitors shrink because the switching frequency can be raised [3], the voltage step at the windings is reduced [9], and a faulty submodule can be bypassed [3]. Analysis of the dc-link stability [5], control [2], [9], [10], machine design [7], [8] and an integrated prototype [6] have been reported; a comparative evaluation for integrated motor

drives found the SPB with 650 V GaN transistors competitive with a three-level flying-capacitor inverter [20], and a GaN-based SPB was shown to reduce the switching losses from 3.2 % to 0.5 % of the output power compared with silicon MOSFETs at 100 kHz [4].

The building block of an SPB, however, is a demanding power stage in its own right. At a phase current of 100 A_{rms} a single low-voltage GaN die is not sufficient, and paralleled GaN half bridges are known to fail by drain–source overshoot [11], by uneven dynamic current sharing caused by asymmetric loops [12], [13] and by gate-voltage disturbances [13], [14]. Low-voltage paralleled-GaN inverters have been demonstrated with symmetric “butterfly” layouts [15], eight parallel devices on a symmetric gate board [16] and hybrid insulated-metal-substrate packaging [13]; efficiency–density co-optimization has been formalized for a 100 kW GaN inverter [17]. What is missing for the SPB is a design procedure that starts from the inverter specification, in which the submodule voltage, the transistor, the number of paralleled devices and the dc link follow from each other instead of being fixed in advance.

This paper proposes such a procedure (Fig. 2) and applies it to a 1200 V, 150 kW SPB. Its contributions are:

- 1) a voltage-class selection showing that, at a given bus voltage and phase current, the minimum semiconductor loss of an SPB scales with the device time constant $R_{\text{DS(on)}}C_{\text{oss,tr}}$, which favors 100 V GaN transistors;
- 2) a layout and gate-drive concept with one star-connected driver per pair of paralleled transistors in two mirrored commutation cells, from which the number of paralleled devices is chosen by loss and auxiliary-circuit count;
- 3) a dc-link design with MLCC hierarchy, ESR/ESL-based current sharing and capacitor losses, and a frequency-domain board copper-loss model showing that the switching harmonics add about one third to the dc copper loss;
- 4) a criterion for when the overlap loss formula $V^2 I / (2 dv/dt)$ applies to a GaN turn-off, and a conduction thermal model that identifies the board copper, rather than the devices, as the limiting thermal path.

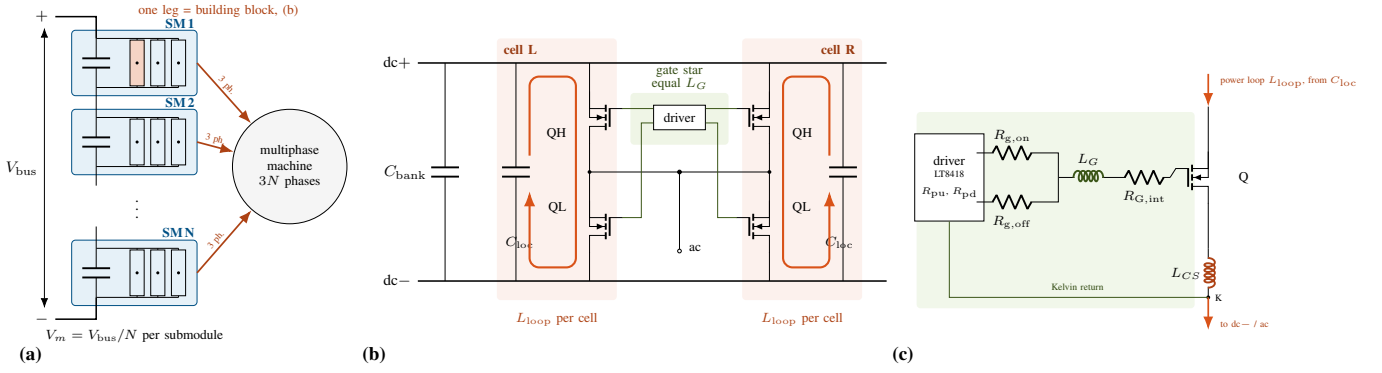


Fig. 1. (a) Stacked polyphase bridge (SPB) inverter: N three-phase submodules (SM) are connected in series on the dc side and feed the N three-phase winding sets of a multiphase machine; each submodule operates at $V_m = V_{bus}/N$. (b) Building block designed in this paper, i.e., one phase leg of a submodule: two mirror-symmetric commutation cells (red) with one high-side (QH) and one low-side (QL) EPC2361 each, so that two transistors are paralleled per switch, local MLCCs C_{loc} in every cell, a common MLCC bank C_{bank} and a centre gate driver (green) with equal, star-connected gate loops. The red arrows indicate the commutation loop of each cell. (c) Gate and power circuit of one transistor: driver pull-up and pull-down resistances R_{pu} , R_{pd} , external gate resistors $R_{g,on}$ and $R_{g,off}$, gate-loop inductance L_G , internal gate resistance $R_{G,int}$, and the common-source inductance L_{CS} shared by the gate loop and the power loop up to the Kelvin connection K in the source pad. The values are given in Table IV.

II. SPB INVERTER AND DESIGN PROCEDURE

The target is a 1200 V, 150 kW traction inverter (Fig. 1a). If the bus voltage V_{bus} is divided into N submodules of $V_m = V_{bus}/N$ that share the power equally, the power of one phase leg at modulation index M and unity power factor is

$$P_{leg} = \frac{M V_m}{2\sqrt{2}} I_{ph}, \quad P = 3N P_{leg} = \frac{3M V_{bus}}{2\sqrt{2}} I_{ph}, \quad (1)$$

so that the phase current I_{ph} is set by the inverter power and the bus voltage alone and is the same for every choice of N . With space-vector modulation ($M = 1.15$), 150 kW at 1200 V requires about 100 A_{rms} per phase. The phase current is therefore the fixed specification of the building block, whereas its voltage, the transistor and the number of submodules remain open and are chosen in the procedure of Fig. 2: the voltage class and the transistor first (Section III), then the layout and gate-drive concept, which determines how transistors can be paralleled (Section IV), the number of paralleled devices (Section V) and the dc link (Section VI); the parasitics, switching energies, losses and temperatures are then verified with one parametric geometry that drives the PCB placement, an Ansys Q3D extraction, LTspice, a frequency-domain copper-loss model and an Ansys Icepak model (Sections VII–X).

III. VOLTAGE CLASS AND TRANSISTOR

The submodule voltage, and with it the transistor voltage class, is the first design decision. Every submodule carries the same phase current I independent of N , see (1). With n transistors per switch, the conduction loss of all submodules scales with $NI^2 R_{DS(on)}/n$ and the loss associated with the output charge with $Nnf_{sw}Q_{oss}V_m$. With $Q_{oss} \approx C_{oss,tr}V_m$ and an optimal number of devices, $n^* = I\sqrt{R_{DS(on)}/(f_{sw}Q_{oss}V_m)}$, the sum of the two over all three legs of all submodules becomes

$$P_{min} = 6 I V_{bus} \sqrt{f_{sw} R_{DS(on)} C_{oss,tr}}. \quad (2)$$

TABLE I
OFF-THE-SHELF GAN TRANSISTORS FOR A 1200 V SPB (100 A_{rms}, 50 kHz)

Device	V_{DS} [V]	$R_{DS(on)}$ [mΩ]	$C_{oss,tr}$ [pF]	V_m [V]	τ [ps]	N	P_{min} rel.
EPC2361	100	0.75	1800	75	1.35	16.0	1.0
IGC033S10S1	100	2.4	860	75	2.06	16.0	1.2
EPC2302	100	1.4	1700	75	2.38	16.0	1.3
EPC2071	100	1.7	1420	75	2.41	16.0	1.3
EPC2215	200	6	690	150	4.14	8.0	1.8
EPC2304	200	3.5	1200	150	4.20	8.0	1.8
EPC2034C	200	6	960	150	5.76	8.0	2.1
EPC2050	350	55	125	240	6.88	5.0	2.3
IGT65R140D2	650	140	35	400	4.90	3.0	1.9
IGT65R025D2	650	25	205	400	5.12	3.0	1.9

To first order the voltage class cancels, and candidates are ranked by the device time constant $\tau = R_{DS(on)}C_{oss,tr}$, a hard-switching device figure of merit (D-FOM) for the SPB.

Fig. 3 and Table I compare ten EPC and Infineon devices from the 100, 200, 350 and 650 V classes, each operated at a submodule voltage of 60–75 % of its rating, i.e., with 16, 8, 5 and 3 submodules. The 100 V devices reach $\tau = 1.35$ –2.4 ps, the 200 V devices 4.1–5.8 ps, the only available 350 V device 6.9 ps and the 650 V devices about 4.9 ps. An SPB built from EPC2361 therefore has a 1.8 times lower minimum loss than one built from the best 200 V device and a 1.9 times lower one than with 650 V devices. Three practical arguments point the same way. First, at 100 A per submodule the optimum number of high-voltage devices per switch becomes impractical (12 for IGT65R025D2), whereas EPC2361 needs about 3.8; the overlap and copper losses, not included in (2), reduce this further (Section V). Second, the 3×5 mm packages of the 100 V class allow sub-nanohenry commutation loops (Section VII). Third, the on-resistance of the 650 V devices rises by a factor

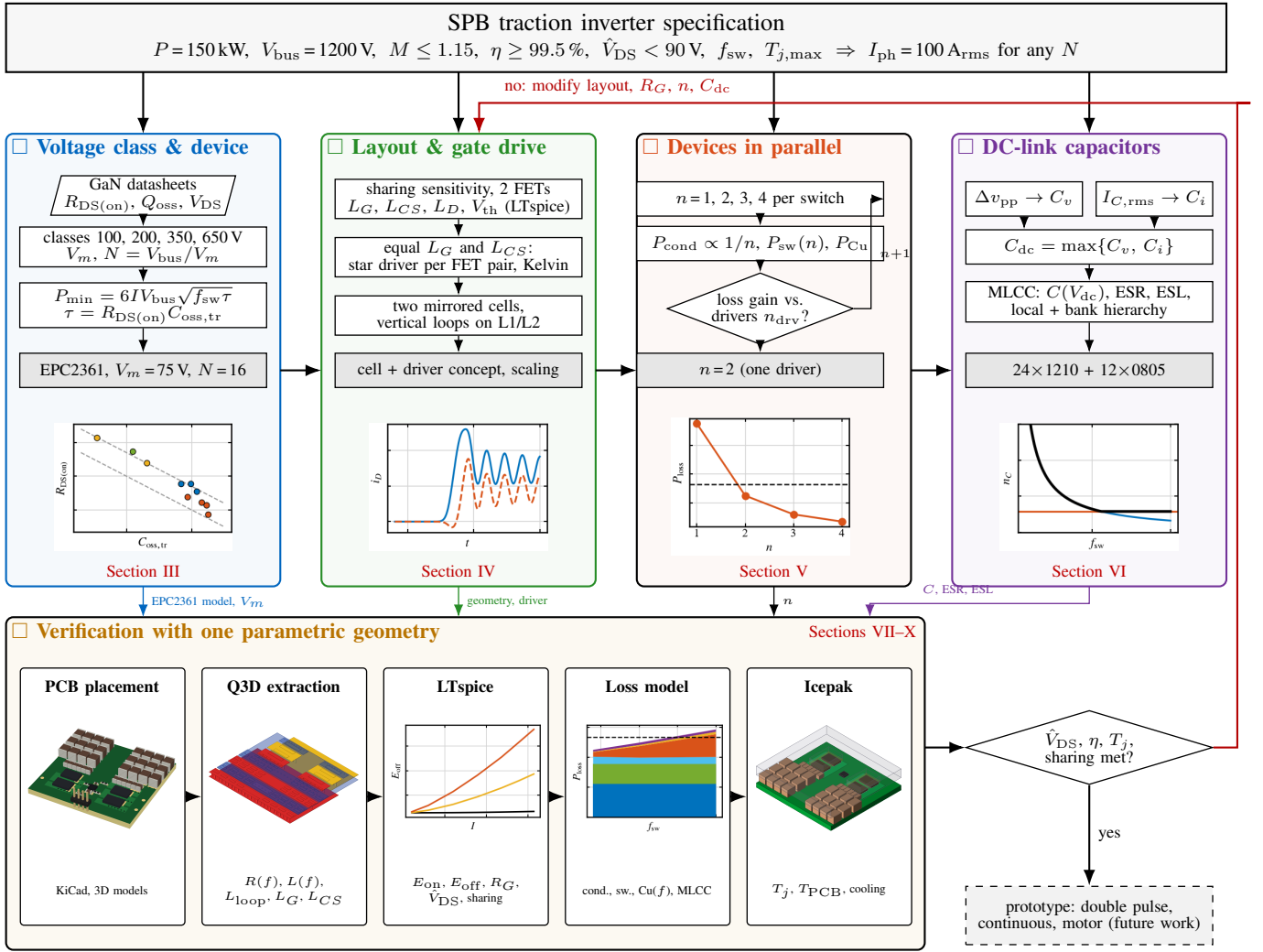


Fig. 2. Design procedure of the SPB building block. From the inverter specification, four selection steps fix the voltage class and transistor (Section III), the layout and gate-drive concept (Section IV), the number of paralleled devices (Section V) and the dc-link capacitors (Section VI). One parametric geometry then drives the PCB placement, the Q3D extraction, the LTspice switching model, the loss model with frequency-dependent copper resistance and the Icepak model (Sections VII–X); if the overshoot, efficiency, temperature or sharing targets are missed, the layout, gate resistors, device count or dc link are modified. The thumbnails show results of the respective steps.

of 2.1 at 150°C , against 1.8 for EPC2361. The price is a larger number of submodules, i.e., more gate drivers, auxiliary supplies and control channels, which the modularity of the SPB accommodates [3]. This contrasts with the 650 V GaN SPB of [20], where the submodule count rather than the device figure of merit was minimized. The 100 V class and EPC2361 (0.75 m Ω typical, 3×5 mm) are therefore selected.

With 100 V transistors, 16 submodules give a nominal submodule voltage of 75 V. The submodule voltage may rise to 90 V with battery-voltage variation and unbalance between submodules, which leaves 10 V to the rating for overshoot and ringing; the peak drain–source voltage must therefore stay below 90 V at the maximum current. From (1), one leg delivers 2.65 kW at $M=1$ and 3.05 kW at $M=1.15$, i.e., about 9.2 kW per submodule and 146 kW for the inverter. Table II summarizes the resulting specification of the building block together with the design decisions and results of the

following sections. Using the device power utilization of [16], $\text{DPU} = V_{\text{dc}}I_{\text{ph}}R_{\text{DS(on)}}/(nV_{\text{DS}}^2) \cdot 10^4$, the block is rated at a DPU of 3.75, comparable to the continuously operated paralleled-GaN inverters reviewed in [16] (2.3–9.5).

IV. LAYOUT AND GATE-DRIVE CONCEPT

Before the number of paralleled devices can be chosen, the way in which transistors are paralleled must be fixed, because it determines both the current sharing and the auxiliary circuits. Paralleled GaN transistors share the conduction loss well, since the on-resistance has a positive temperature coefficient, but not necessarily the switching loss: a transistor whose gate loop is longer, or whose common-source inductance differs, turns on later and off earlier than its neighbor and takes a larger part of the switching energy [12], [13]. How strongly the layout asymmetry matters is quantified with an LTspice double-pulse model of two paralleled EPC2361 at

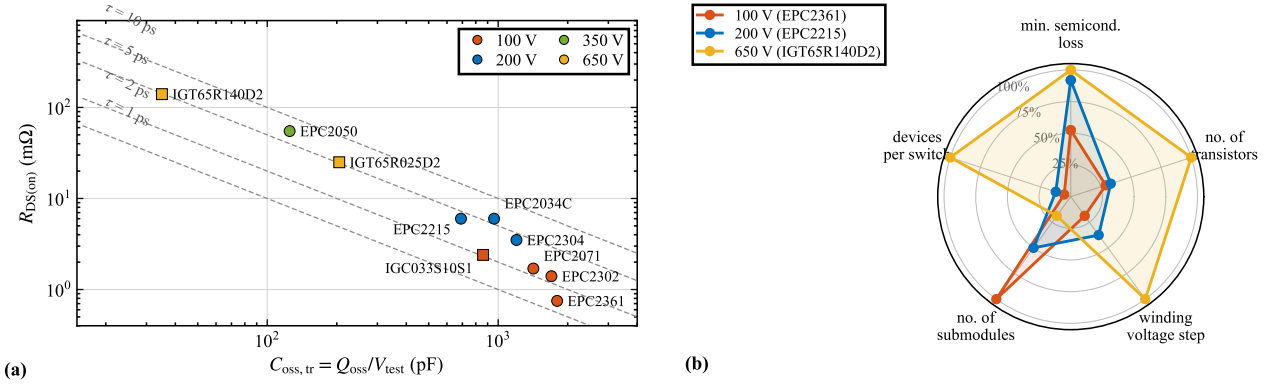


Fig. 3. Voltage-class selection for a 1200 V SPB with 100 A_{rms} phase current at 50 kHz. (a) Typical on-resistance at 25 °C against the time-related output capacitance $C_{oss,tr} = Q_{oss}/V_{test}$ of off-the-shelf GaN transistors (circles: EPC, squares: Infineon; datasheet values), with lines of constant device time constant $\tau = R_{DS(on)}C_{oss,tr}$. (b) Comparison of SPBs built from the best device of the 100, 200 and 650 V classes, each axis normalized to its maximum: minimum conduction plus output-charge loss (2), optimum number of devices per switch n^* , number of submodules N , voltage step at the winding V_m and total number of transistors $6Nn^*$.

TABLE II
BUILDING-BLOCK SPECIFICATION, DESIGN AND KEY RESULTS

Specification (1200 V, 150 kW SPB, $N = 16$)	
Submodule voltage nominal / max.	75 V / 90 V
Phase current	100 A _{rms} (141 A peak)
Power per leg ($M = 1 / 1.15$)	2.65 / 3.05 kW
Targets	$\eta \geq 99.5\%$, $\hat{V}_{DS} < 90$ V at 160 A
Design	
Transistors	2 × EPC2361 (100 V, 0.75 mΩ) per switch
Gate drive	one half-bridge driver, star-connected gates
Board	30 × 28.2 mm, 4 layers, 2 oz, 0.1 mm L1–L2
DC link per leg	24 × 10 μF 1210 + 12 × 1 μF 0805 MLCC
$R_{g,on}/R_{g,off}$ (external)	2/0 Ω (3.0/0.8 Ω total)
Results (simulation)	
Commutation loop per cell (Q3D)	0.312 nH copper, 0.374 nH with ESL
Gate loop per transistor (Q3D)	1.70 nH, equal for all four
Peak V_{DS} at 160 A (HS/LS)	83.1 / 85.1 V
Loss / efficiency at 50 kHz	12.49 W / 99.53 %
T_j at 60 °C cold plate	72 °C (67 °C with bottom pad)

75 V and 120 A, in which one asymmetry is introduced at a time (Fig. 4). With a symmetric layout and Kelvin source connections both transistors carry the same current and energy. A gate-loop mismatch of 2 versus 8 nH delays the second gate by about 2 ns, half of the 4 ns current rise, and gives a dynamic imbalance of 19 % with a 4:1 split of the turn-on energy (Fig. 4a.ii). Without Kelvin connections, a symmetric common-source inductance only slows both transistors, but a 0.2 nH mismatch of it is the worst case, with 29 % imbalance and 2.8 times the turn-on energy in one device (a.iii). A threshold-voltage spread of 0.3 V gives 7 % at turn-on but a 2:1 split of the turn-off energy, and a 0.3 nH drain-path mismatch only 2 %. The static sharing at the end of the pulse is 59–60 A per device in every case, since it is set by $R_{DS(on)}$. Asymmetric switching therefore concentrates the switching loss in one device, which then sets the thermal limit of the whole switch; the layout must make the gate loops and the common-source paths equal, whereas the power-loop paths are less critical.

The building block therefore uses one half-bridge gate driver (LT8418) for every pair of paralleled transistors, placed on the symmetry line between them and connected to both gates in a star with Kelvin source returns, so that both gate loops are equal by construction. The power stage is split accordingly into two mirror-symmetric commutation cells, each with one high-side and one low-side transistor and its own local capacitors (Fig. 1b). The concept scales in pairs (Fig. 6a): one driver serves one or two devices per switch; three devices per switch need a second driver, since a third branch cannot be made equal to the other two, and four devices per switch are built as two of the two-device structures with two drivers. The number of drivers, and with them the isolated auxiliary supplies, level shifters and control channels, thus doubles from two to three devices per switch.

In each cell the commutation loop is vertical: the forward path on layer L1 is closed on the dc– plane on L2, 0.1 mm below, so that the opposing currents cancel their flux. The dc+ plane on L3 and the ac plane on L4 carry the load current to the terminals on the symmetry line, and the capacitor bank above the cells feeds both cells equally, which avoids the shared, cascaded loop inductance identified in [11]. The driver sits in the 6 mm corridor between the cells. The board measures 30 × 28.2 mm (8.5 cm², 0.31 kW/cm² at board level). The same parametric geometry generates the KiCad placement, the Q3D model and the Icepak model (Fig. 5), so that a change of, e.g., the prepreg thickness propagates to the inductances, losses and temperatures without manual steps.

V. NUMBER OF DEVICES IN PARALLEL

The number of transistors per switch n trades the conduction loss, which falls with $1/n$, against the current-independent part of the switching energy, which grows with n , and against the number of drivers. Fig. 6(b) evaluates the leg loss with the switching energies of Section VIII and the copper and capacitor losses of Sections VI and IX. One device per switch misses the efficiency target at all frequencies (17.5 W

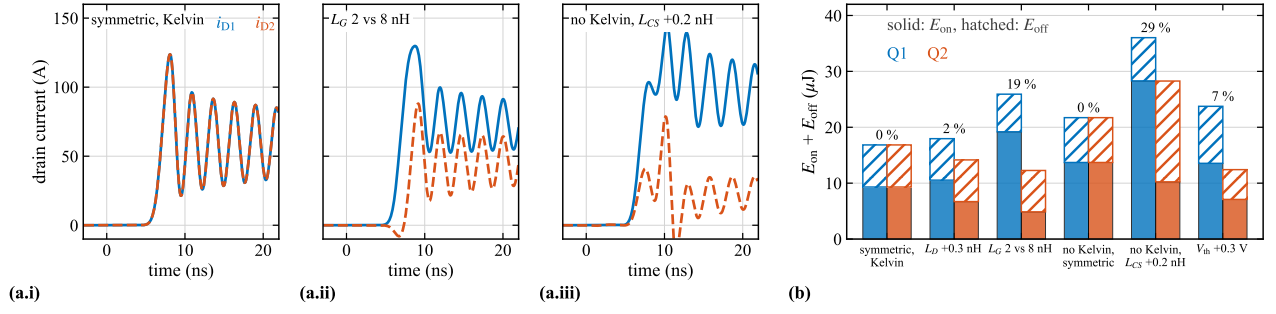


Fig. 4. Sensitivity of the current sharing of two paralleled EPC2361 (LTspice, low-side double pulse, 75 V, 120 A, $R_G = 2 \Omega$, baseline $L_G = 2$ nH, $L_D = 0.15$ nH, $L_S = 0.05$ nH). (a) Drain currents at turn-on for (a.i) the symmetric layout with Kelvin connections, (a.ii) a gate-loop mismatch of 2 versus 8 nH and (a.iii) no Kelvin connection with a common-source inductance mismatch of 0.2 nH. (b) Turn-on (solid) and turn-off (hatched) energy of the two transistors for one asymmetry at a time, with the dynamic current imbalance $(i_1 - i_2)/(i_1 + i_2)$ at the turn-on peak.

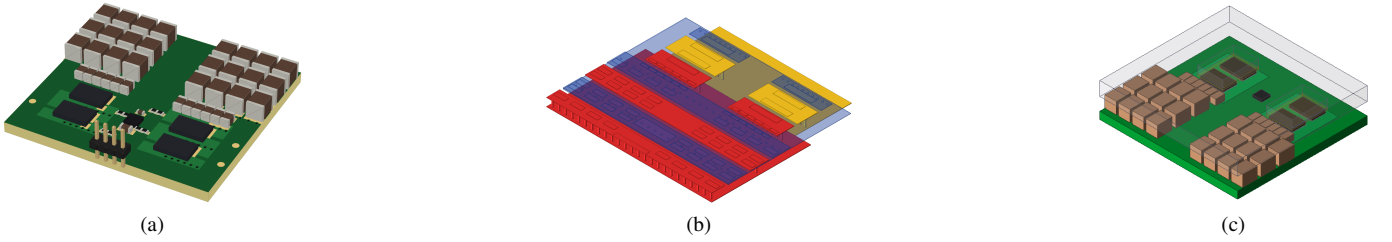


Fig. 5. One parametric geometry, three models. (a) KiCad placement with the component 3D models (30×28.2 mm, four layers, 2 oz copper): MLCC bank (top), local MLCCs, the two mirrored cells with QH and QL, and the centre driver. (b) Ansys Q3D copper model with the dc+ (red), dc- (blue, transparent) and ac (yellow) nets and the via rows of the vertical loops. (c) Ansys Icepak model with gap pads, pedestals and the aluminium cold plate (shown above the board).

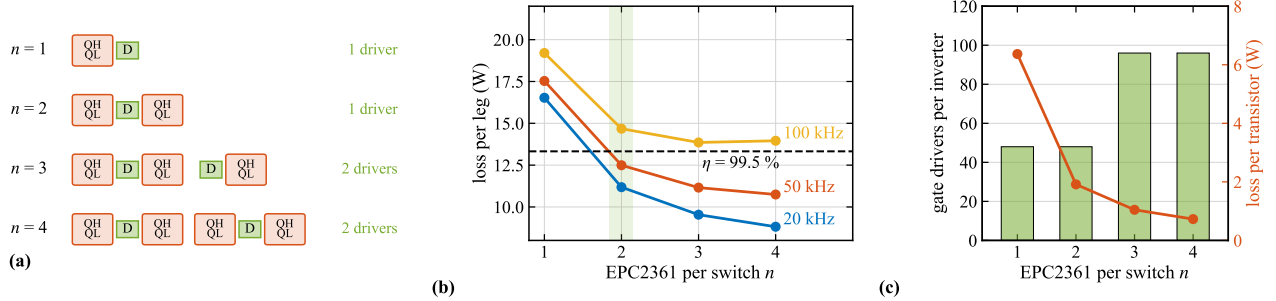


Fig. 6. Number of EPC2361 per switch. (a) Scaling of the layout concept: two-device cells (red) with one star-connected half-bridge driver (D, green) per pair; three devices per switch need a second driver, four are built from two two-device structures. (b) Loss per leg at the rated point (75 V, $100 A_{rms}$, $M = 1$, typical $R_{DS(on)}$ at $100^\circ C$, including copper and capacitor losses) for 20, 50 and 100 kHz; the dashed line is the 99.5% target. (c) Gate drivers of the complete 1200 V inverter (16 submodules, 48 legs) and loss per transistor at 50 kHz.

at 50 kHz, 6.4 W per device, which is also difficult to cool through a 3×5 mm package). The step from one to two devices removes 11.1–5.5 W of conduction loss and reduces the leg loss to 12.5 W (99.53%) and the device loss to 1.9 W. Three and four devices reduce the leg loss by only a further 1.3 and 1.7 W, because the conduction loss is already smaller than the copper loss and the output-charge loss grows (1.9, 2.4 and 2.9 W switching loss for two, three and four devices). At the same time, three devices per switch double the number of gate drivers from 48 to 96 in the complete inverter (Fig. 6c), together with their isolated supplies and signal channels, and lengthen the loops. Two devices per switch, i.e., two cells with one centre driver, are therefore selected.

VI. DC-LINK CAPACITOR DESIGN

Each submodule needs its own dc link, which must (i) limit the switching-frequency voltage ripple, (ii) carry the ripple current of the legs, (iii) close the commutation loops with a minimum of inductance and (iv) fit on the building block. With three legs on one submodule bus, the high-frequency capacitor current of one leg is $28.8 A_{rms}$ at the rated point and up to $50 A_{rms}$ at low modulation index and zero power factor; limiting the peak-to-peak ripple to 4% (3 V) at 50 kHz requires an effective capacitance of $78 \mu F$ per leg, compared with $236 \mu F$ if one leg had to supply its own ripple.

Film capacitors avoid the dc-bias loss of capacitance but need several times the volume at 100 V and add lead in-

TABLE III
DC-LINK CAPACITORS PER LEG (75 V, 50 KHz, RATED POINT)

	local (per cell)	bank
Part	TDK C2012X7S2A105K125AB	Murata GRM32EC72A106KE05L
Size, dielectric	0805, X7S, 100 V	1210, X7S, 100 V
Number per leg	2 × 6	24 (3 rows of 8)
C_{nom} / C_{eff} at 75 V*	1 μ F / 0.45 μ F	10 μ F / 3.0 μ F
ESR* / ESL* per part	6 m Ω / 0.48 nH	3 m Ω / 0.5 nH
Self-resonance	10.8 MHz	4.1 MHz
Ripple current per part (network)	1.1 A _{rms}	2.0–2.1 A _{rms}
Loss per part	7.5 mW	11.5–13.3 mW

*assumed; to be confirmed with manufacturer models and impedance measurements

ductance, and electrolytic or polymer capacitors cannot carry this ripple current per volume at 50 kHz and above. Class-II MLCCs are therefore used, in two levels (Fig. 1b, Table III). Six 1 μ F 0805 capacitors per cell sit directly at the high-side drain and define the commutation loop, and a bank of 24 10 μ F 1210 capacitors in three rows above the cells supplies the ripple charge. Because the capacitance of X7S dielectrics falls strongly with dc bias, the effective values at 75 V are taken as 30 % (1210) and 45 % (0805) of the nominal values, which gives 77 μ F per leg and meets the ripple requirement at 50 kHz. Fig. 7(a) shows that the number of bank capacitors is set by the voltage ripple up to about 98 kHz and by the ripple current per capacitor above, where raising the switching frequency no longer reduces the dc-link size.

The division of the ripple current between the capacitor groups depends on their ESR and ESL and on the board copper between them, so it is computed with the frequency-domain network of Section IX, in which every capacitor group is represented by its C_{eff} , ESR and ESL (Fig. 7b). At the switching frequency the bank is the lower impedance and carries most of the ripple: 2.0–2.1 A_{rms} per 1210 capacitor, the third row slightly more, against 1.1 A_{rms} per local 0805 capacitor (Fig. 7c). Above the self-resonance of the 1210 parts (about 4 MHz) the local capacitors take over and supply the fast commutation transients. The total capacitor loss is 0.39 W per leg at the rated point, small compared with the other losses but concentrated in parts whose ripple current is already about 84 % of the assumed 2.5 A rating; the bank is therefore designed close to both its voltage-ripple and its current limit at 50 kHz.

VII. PARASITIC AND GATE-LOOP DESIGN

A. Power and gate loop

The partial resistance and inductance matrix of the copper is extracted with Ansys Q3D at 100 MHz, every component being represented by source and sink terminals on its pads, and is exported to LTspice as coupled inductors. Exciting both cells simultaneously gives a commutation-loop inductance of 0.312 nH (copper) and 0.374 nH including the capacitor ESL per cell. Each transistor sees this loop; paralleling does not reduce the inductance, but each cell commutates half of the load current, so that the di/dt per loop and thus the overshoot are halved. The value excludes the package inductance and routing details and is therefore a lower bound.

TABLE IV
EXTRACTED PARASITICS AND GATE-DRIVE PARAMETERS

Power loop (Q3D, 100 MHz)		
L_{loop} per cell, copper / with MLCC ESL	BB_PL assumed	0.312/0.374 nH
ESL per MLCC, 1210 / 0805		0.50/0.48 nH
Gate loop (Q3D, 100 MHz)		
L_G per transistor (gate and Kelvin return)	BB_CS	1.71 nH
L_{CS} , Kelvin via in the source pad	BB_CS	0.030 nH
L_{CS} , driver ground on the dc— plane	BB_CS_noK	0.061 nH
Gate loop to power loop of the other cell	BB_CS	0.010 nH
Mutual between the two gate loops	BB_CS	0.46 nH
Gate resistances		
$R_{g,on} / R_{g,off}$ (external)	selected	2/0 Ω
R_{pu} / R_{pd} (driver output)	model	0.6/0.2 Ω
$R_{G,int}$ (EPC2361)	datasheet	0.4 Ω
$R_{G,on} / R_{G,off}$ total per transistor		3.0/0.8 Ω

The dielectric thickness h between L1 and L2 is the dominant geometric parameter. Q3D gives 1.60 nH per millimetre of dielectric, close to the 1.40 nH/mm of the parallel-plate estimate $\mu_0 h l / w$ used in [13], [16] (Fig. 8a), plus a fixed 0.18 nH from vias, pads and package that no dielectric reduction can remove. At 0.1 mm (two-ply prepreg) the fixed part is already half of the loop.

The gate star gives 1.70 nH per transistor. A side-mounted driver, as in a conventional single-row layout, would give 2.0 and 7.7 nH to the near and far transistors (Q3D); the centre driver makes both branches equal by construction.

The common-source inductance L_{CS} is the part of the source path that the gate loop and the power loop share. It is extracted with a combined Q3D model that contains the power loop, the gate star and its return: the Kelvin return leaves each QL source pad through a via and runs on L2 directly under the gate trace, in a slot of the dc— plane, to the driver ground. With every loop described by its terminal currents, the gate-loop inductance is $L_G = \mathbf{g}^T \mathbf{L} \mathbf{g}$ and the common-source inductance is the mutual term $L_{CS} = \mathbf{g}^T \mathbf{L} \mathbf{p}$ between the gate loop $\mathbf{g}$ and the commutation loop $\mathbf{p}$ of the same cell. The Kelvin layout gives $L_{CS} = 0.030$ nH, equal in both cells, and $L_G = 1.71$ nH, which confirms the gate-star extraction; the coupling of a gate loop to the commutation loop of the other cell is only 0.010 nH. At the turn-on di/dt of about 27 A/ns per transistor, L_{CS} still induces about 0.8 V of negative feedback in the gate loop, but equally in both transistors, so that it slows rather than unbalances the switching (Fig. 4). If the driver ground is instead connected to the dc— plane next to the driver, the shared path grows to 0.061 nH. Table IV collects the extracted parasitics and the gate-drive parameters used in the circuit simulations.

B. Gate resistors and current sharing

Treating the gate loop as a series RLC with C_{iss} gives a lower bound for the turn-on resistance, $R_{G,on} \geq 2\zeta \sqrt{L_G / C_{iss}}$, and the Miller current an upper bound for the turn-off resistance, $R_{G,off} \leq V_{th} / (C_{rss} dv/dt)$ [13] (Fig. 8b). Inside this window, an LTspice sweep with the extracted parasitics at 160 A selects external resistors of

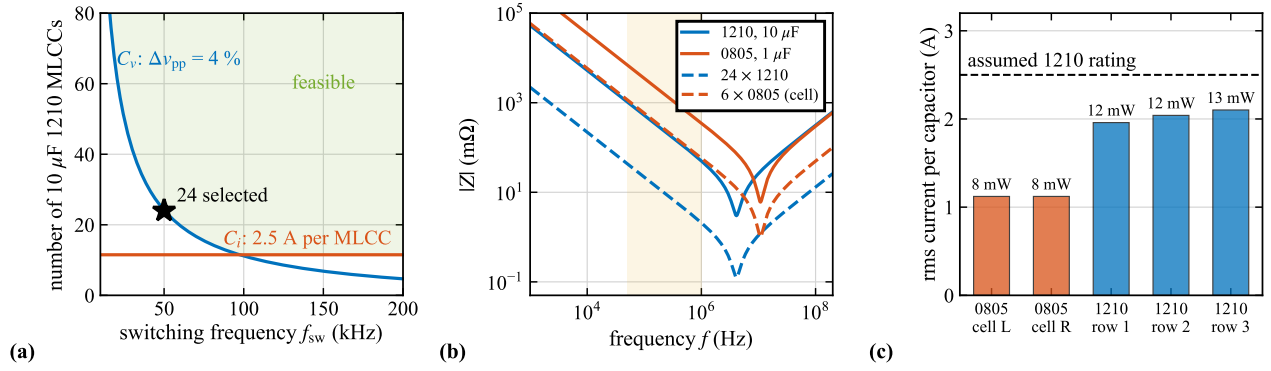


Fig. 7. DC-link design of the building block. (a) Number of 10 μF 1210 MLCCs per leg against switching frequency, required for a 4% peak-to-peak voltage ripple on a three-leg submodule bus (C_v , blue) and for an rms current of 2.5 A per capacitor (C_i , red); the shaded area is feasible, the star marks the selected bank. (b) Impedance magnitude of single capacitors (solid) and of the bank and the local group of one cell (dashed) from the C_{eff} –ESR–ESL model; the shaded band covers the switching frequency and its first harmonics. (c) RMS ripple current and loss per capacitor at the rated point, from the frequency-domain network of the Q3D board impedances and the capacitor models.

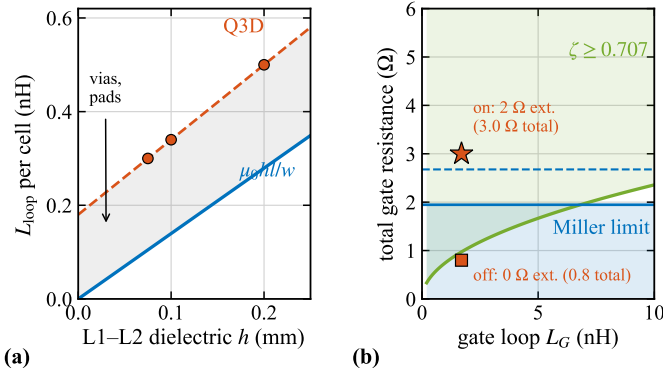


Fig. 8. (a) Commutation-loop inductance per cell against the L1–L2 dielectric thickness h : Q3D (circles, linear fit dashed) and parallel-plate estimate $\mu_0 h l / w$; the shaded difference is the fixed contribution of vias and pads. (b) Gate-resistor window per transistor in terms of the total gate resistance: damping bound at turn-on ($\zeta \geq 0.707$, green) and Miller bound at turn-off (blue, solid for $V_{th,min} = 0.8 \text{ V}$, dashed for 1.1 V); the markers are the selected external 2 Ω (on) and 0 Ω (off) resistors, i.e., 3.0 and 0.8 Ω in total, at $L_G = 1.70 \text{ nH}$.

$R_{g,on}/R_{g,off} = 2/0 \Omega$, i.e., 3.0/0.8 Ω in total with the 0.4 Ω internal gate resistance and the 0.6/0.2 Ω driver pull-up/pull-down (the pull-down shared by the two transistors): below 2 Ω the high-side overshoot exceeds 90 V (112 V at 0.5 Ω), above it E_{on} rises. The peak V_{DS} is 83.1 V and 85.1 V for the high and low side, and the switching speeds are 12 and 32 V/ns at turn-on and turn-off. The off-state bump of the high-side gate is 0.98 V; it exceeds the minimum threshold of 0.8 V and is the tightest margin of the design.

Applied to the complete building block with the extracted parasitics, the symmetric layout of Section IV (Fig. 4) leaves the two paralleled transistors with 78.53 and 78.63 A at turn-off, i.e., a dynamic imbalance below 0.1%.

VIII. SWITCHING ENERGIES AND SLOW SWITCHING

With the selected resistors, one switch position dissipates $E_{on} = 43 \mu\text{J}$ and $E_{off} = 8.1 \mu\text{J}$ at 141 A. The turn-on energy

consists of a current-independent part of $12 \mu\text{J}$, of the order of $Q_{oss} V$, and an overlap part that follows $V^2 I / (2 dv/dt)$ within a few percent. The turn-off energy is almost independent of current: the gate is discharged through about 0.8Ω faster than the load current can move the output charge, so the channel is off before V_{DS} rises, the voltage rises at $dv/dt \approx I / C_{oss,tot}$ (5.3 nF), and most of the 7–8 μJ measured at the device terminals is energy stored in C_{oss} .

A motor-friendly design may slow the switching with larger gate resistors. Fig. 9 compares six resistor pairs simulated with the full building-block model. The turn-off becomes channel-controlled only when the gate removes the Miller charge more slowly than the load current moves the output charge, i.e. for

$$R_{g,off,tot} > R_{crit} = \frac{V_{pl} Q_{oss,tot}}{I Q_{GD}}, \quad (3)$$

which gives about 1.5 Ω at 139 A and 7 Ω at 30 A ($V_{pl} \approx 2 \text{ V}$, $Q_{GD} = 3.8 \text{ nC}$). Above R_{crit} the voltage term of the classical overlap formula predicts E_{off} within about 10%, whereas the additional current-fall term $I^2 V / (2 di/dt)$ overestimates it by a factor of 1.8; below R_{crit} the formula overestimates the loss by a factor of two. At turn-on, the di/dt term is needed only when the gate resistance makes the current rise a separate, slow phase (10 Ω).

Two consequences follow for the drive design. First, $R_{g,on} = 10 \Omega$ mainly slows the current rise: the turn-on dv/dt falls only from 13 to 11 V/ns while E_{on} rises 2.5 times; the turn-off dv/dt is controlled by $R_{g,off}$ only above R_{crit} . Second, the same $R_{g,off}$ holds the complementary transistor off, and its gate bump rises from 0.99 to 1.29 V; slowing the turn-off therefore requires a separate low-impedance hold-off path. At 50 kHz the leg efficiency falls from 99.53% for the selected 2/0 Ω pair to 99.50, 99.45 and 99.38% for 2/5, 10/2 and 10/10 Ω, so that the fully slowed design misses the efficiency target.

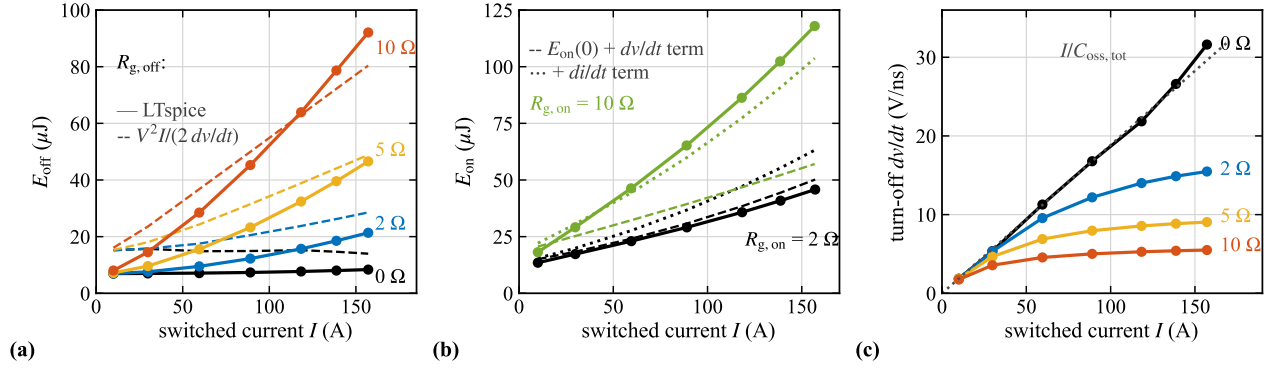


Fig. 9. Switching energies per switch position of the building block with slow switching (LTspice with the Q3D parasitics, 75 V, $R_{g,\text{on}}$ and $R_{g,\text{off}}$ external resistances). (a) Turn-off energy (solid) and voltage-overlap estimate $V^2 I / (2 dv/dt)$ with the simulated dv/dt (dashed) for $R_{g,\text{off}} = 0, 2, 5$ and 10Ω ; the estimate fails below the critical resistance (3). (b) Turn-on energy for $R_{g,\text{on}} = 2$ and 10Ω against the current-independent part $E_{\text{on}}(0)$ plus the voltage-overlap term (dashed) and plus the current-overlap term (dotted). (c) Turn-off dv/dt against the load-driven limit $I/C_{\text{oss,tot}}$ (dotted): without external resistance the voltage rise is set by the load current, with 5–10 Ω by the gate.

IX. LOSSES INCLUDING FREQUENCY-DEPENDENT PCB COPPER

The leg loss is evaluated for a sinusoidal current with continuous pulse-width modulation, for which conduction, switching and dead-time losses are identical for sinusoidal and space-vector modulation. With synchronous conduction, the high- and low-side transistors also share the loss equally for any modulation index and power factor, since $\overline{di^2} = \overline{i^2}/2$.

The PCB copper loss is usually computed with dc resistances. The dc-link current of a phase leg, however, contains about $29 I_{\text{rms}}$ at the switching frequency and its harmonics, which circulates between the MLCC bank and the cells. A Q3D frequency sweep from 1 kHz to 100 MHz shows that the branch resistances are already 1.5 times their dc value at 50 kHz and 3 times at 500 kHz (Fig. 10a), because the current crowds onto the facing plane surfaces and the lowest-inductance paths. The board is therefore solved as a frequency-domain network of the Q3D $R(f)$ and $L(f)$ matrices, the capacitor groups and the dc-supply impedance, excited by the switched transistor currents at every harmonic of the modulated waveforms:

$$P_{\text{Cu}} = \sum_h \frac{1}{2} \mathbf{I}_h^H \mathbf{R}(f_h) \mathbf{I}_h. \quad (4)$$

The load-frequency part reproduces the dc estimate (3.30 W); the switching harmonics add 1.07 W at the rated point and up to 2.5 W at low modulation index, where the ripple current is largest. The same network gives an MLCC loss of 0.39 W.

Fig. 10(b) and (c) show the resulting loss breakdown. At 50 kHz the leg dissipates 12.49 W (99.53 %): 5.55 W conduction, 4.37 W copper, 1.91 W switching, 0.20 W dead time and 0.39 W in the capacitors. Conduction and copper account for about 80 % of the loss, so cooling and copper design matter more than switching speed at this frequency, which is why the gate resistors could be selected for robustness. The 99.5 % target holds up to about 60 kHz; at 100 kHz the efficiency is 99.45 %, and with the maximum on-resistance the target is

missed already at 50 kHz (99.46 %). Without the switching-harmonic copper term, the same design would have appeared to reach 99.50 % at 100 kHz.

The dc link of Section VI adds 0.39 W, and its size, not the transistors, bounds the useful switching frequency from the density point of view above about 98 kHz.

X. THERMAL DESIGN

A steady-state conduction model is built in Ansys Icepak from the same geometry. The board is an anisotropic equivalent of FR-4 and copper (in-plane 55 W/mK, through-plane 0.36 W/mK, 3.5 W/mK in the cell regions with via arrays); each transistor is a 1.9 W source with a junction-to-case resistance of 0.2 K/W to the top and a junction-to-board resistance of 1.5 K/W; a 0.5 mm gap pad (1.9 K/W per device) and aluminium pedestals connect the transistors to a 3 mm aluminium plate held at 60 °C (liquid cold plate). The board copper loss (4.4 W), the capacitor losses and the driver loss are applied as volumetric sources.

With the cold plate on the transistors only, the junctions stay at 72 °C, 7 K above a per-device estimate, because the transistors also conduct the heat of the board. The board itself, however, reaches 104 °C in the capacitor bank, where the 1210 capacitors reach 103 °C and the driver 91 °C (Fig. 11): its copper loss, larger than the loss of each transistor, has no path to the cold plate other than laterally through the board and upwards through the transistors. A bottom-side gap pad to a second plate lowers the board maximum to 83 °C and the junctions to 67 °C. Since the model distributes the copper loss over the board thickness and neglects convection, the absolute board temperatures are conservative; the conclusion that a low-voltage, high-current building block needs a cooling path for its board copper is not.

XI. DISCUSSION AND FUTURE WORK

Compared with the reviewed paralleled-GaN half bridges (Table V), whose commutation loops range from 0.34 to 6.4 nH, and up to 11 nH in a commercial module [11], the

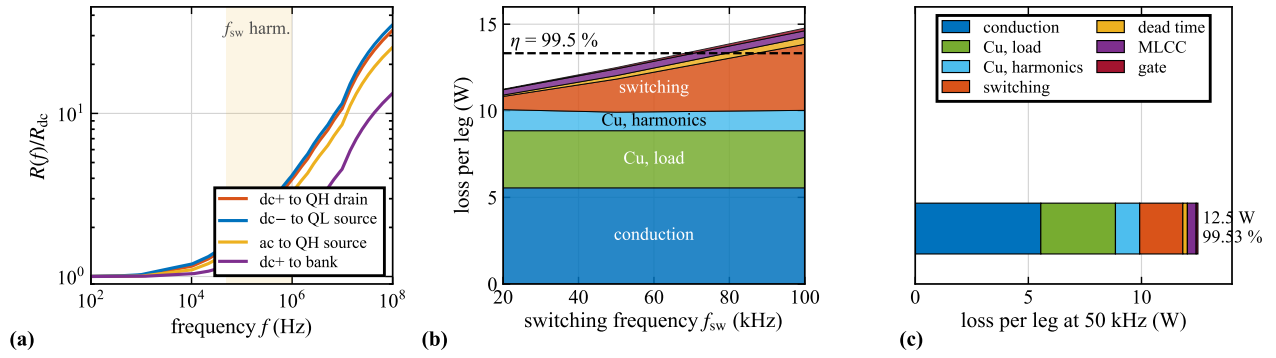


Fig. 10. Losses of the building block at the rated point (75 V, 100 A_{rms}, $M=1$, typical $R_{DS(on)}$ at 100 °C). (a) Q3D resistance of the main board branches against frequency, normalized to the dc value; the shaded band covers the switching frequency and its first harmonics. (b) Loss per leg against switching frequency with the efficiency target. (c) Loss breakdown at 50 kHz; conduction and copper loss together account for about 80 %.

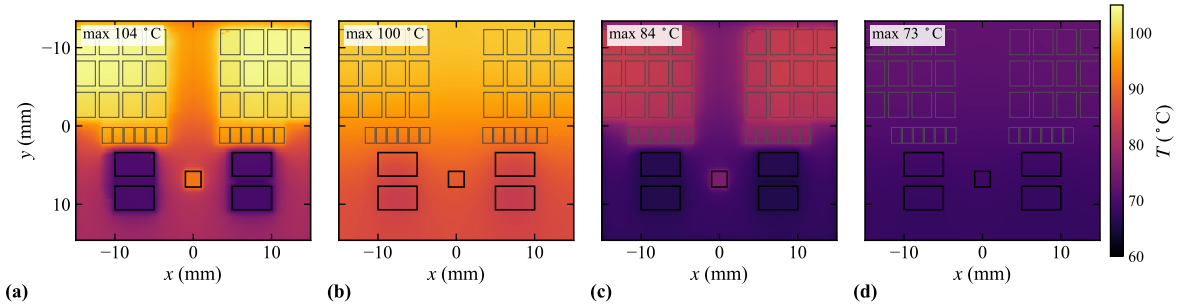


Fig. 11. Icepak temperature maps at the rated point (50 kHz) in the device layer ($z=2.0$ mm, including the air between components) and the board mid-plane. (a), (b) Cold plate on the transistors only. (c), (d) Additional bottom-side gap pad (3 W/mK) to a second plate at 60 °C. Outlines: transistors, MLCCs and driver.

TABLE V
COMPARISON WITH PARALLELED-GAN HALF BRIDGES

Work	Devices/switch	Rating / test	L_{loop}	Layout
[12]	4 × 650 V	400 V/240 A DPT	0.7 nH	6-layer PCB
[13]	4 × 100 V	48 V/328 A DPT	1.14 nH	IMS hybrid
[16]	8 × 200 V	120 V/155 A, 100 kHz	0.34 nH*	PCB, symm. gate
[15]	8 × 100 V	48 V/125 A _{rms}	–	butterfly PCB
[19]	4 × 650 V	350 A _{rms} WLTP	3.9 nH	PCB module
[17]	3 × 650 V	400 V, 100 kW	6.4 nH	PCB + Cu blocks
This work	2 × 100 V	75 V/141 A (design)	0.37 nH/cell	mirrored cells

*analytical estimate

proposed block reaches 0.374 nH per cell with two loops in parallel, and its symmetric gate drive removes the dominant sharing mechanisms. The analysis identifies four limits that guide the next iteration: the off-state gate bump, which calls for a Miller clamp, a small negative off-voltage or a gate-source capacitor [15]; the copper loss, which can be reduced by about 2 W with busbar contacts next to each cell and heavier inner layers, and further by placing more capacitance next to the cells to shorten the ripple path; the thermal path of the board; and the dc-link ripple on a shared submodule bus, which is at its limit.

All results in this paper are from analysis, finite-element extraction and circuit simulation. Experimental validation is future work: double-pulse tests at 75 V and up to 160 A, in-

cluding the loop inductance from the ringing frequency and the per-transistor currents, continuous operation for efficiency and thermal-image based sharing checks [16], measurement of the capacitor impedances, and finally an SPB submodule driving a multiphase machine, where the trade-off between inverter and machine losses will set the switching frequency [21].

XII. CONCLUSION

A design procedure for the building block of stacked polyphase bridge converters has been presented and applied to a 1200 V, 150 kW traction inverter. The device time constant $R_{DS(on)}C_{oss,tr}$ selects the 100 V class and EPC2361, hence 16 submodules of 75 V; a star-connected driver per pair of paralleled transistors fixes the layout; and the loss gain from one to two devices per switch, compared with the small gain and the doubled driver count for three, sets two devices per switch. The dc link of 24 1210 and 12 0805 MLCCs per leg is limited by voltage ripple and ripple current at about the same frequency. The block reaches 0.374 nH per cell and 99.53 % efficiency at 50 kHz on 8.5 cm². The design shows that the switching harmonics of the dc-link current add about one third to the dc copper loss of the board, that the overlap loss formula applies to a GaN turn-off only above the critical gate resistance (3), and that the board copper, rather than the transistors, sets the thermal design of a low-voltage, high-current building block.

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